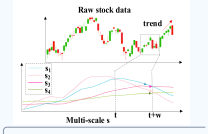
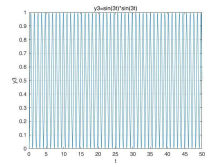
MTDNN

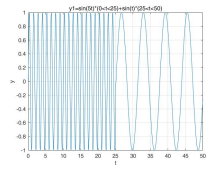
研究背景

- 问题定义
  - 股票趋势预测（STP，Stock Trend Prediction）
  - 输入
    - 股票历史数据序列
    - 开盘价序列、收盘价序列、成交量序列、成交额序列等
  - 输出：股价变动趋势，分为上涨、下跌、不变三类
  - $y_k = \begin{cases} -1 & \Delta p_T \le -\alpha \\ 0 & -\alpha < \Delta p_T < \alpha \\ 1 & \Delta p_T \ge \alpha \end{cases}$　　$\Delta p_T = \dfrac{m_T(k) - p_T}{p_T}$
- 动机
  - 现有的研究大部区考虑单一尺度下的信息，导致了信息的丢失

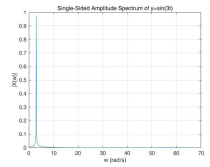


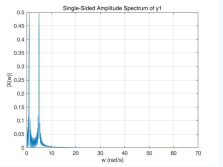
  - 举例
    - 四条折线是对原始信息进行不同尺度的平滑之后得到的，其中只有S4正确表明窗口中时序的走势（简单移动平均，SMA）
- 多尺度信息
  - 1）多种尺度下的股票价格行为信息
    - 不同步长的时域信息
  - 2）多种类型的尺度信息
    - 时域信息
    - 频域信息
    - .....
- 离散小波变换（DWT）
  - 作用：将信号从时域变换到频域
  - 傅里叶变换
    - 时域信号



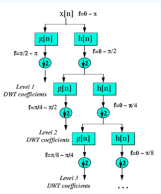


    - 频域信号

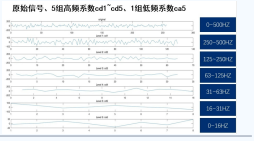




  - 小波变换过程图



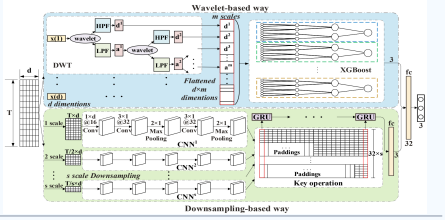
    - g(n)：高通滤波器
    - h(n)：低通滤波器
    - 动态分辨率（海森堡不确定原理）
      - 对于高频信号：窄窗口，获得高的时域分辨率
      - 对于低频信号：宽窗口，牺牲部分时域分辨率以提升频域分辨率



  - 参考资料：从傅里叶变换进到离散小波变换 https://zhuanlan.zhihu.com/p/661 89212

模型介绍

- 模型架构图



- Wavelet-based Way

$$a_n^1 = \sum_t h[2n-t]x_t\,,\qquad d_n^1 = \sum_t g[2n-t]x_t\,,$$

$$a_n^m = \sum_t h[2n-t]a_t^{m-1}\,,\qquad d_n^m = \sum_t g[2n-t]a_t^{m-1}\,,$$

$$\mathbf{v}_i = [\mathbf{d}^m(i), \mathbf{d}^{m-1}(i), ..., \mathbf{d}^1(i), \mathbf{a}^1(i)]\,,$$

- Downsampling-based Way

$$\mathbf{x}^s = \{x_{1+ls}\}_{l\in[0,L^s]}$$

$$v_i = \begin{cases} [u_i^1, \mathbf{0}, ..., \mathbf{0}], & i \in [0, L^1 - \hat{L}^2] \\ [u_i^1, u_{i-L^1+\hat{L}^2}^2, ..., \mathbf{0}], & i \in (L^1 + \hat{L}^2, L^1 + \hat{L}^3] \\ \vdots & \\ [u_i^1, u_{i-L^1+\hat{L}^2}^2, ..., u_{i-L^1+\hat{L}^s}^s], & i \in (L^1 - \hat{L}^S, \hat{L}^1] \end{cases}$$

右对齐，左侧补0

实验与分析

- 数据集

Dataset		Train (%)	Train samples	Test (%)	Test samples
FI-2010	↓	32.03	352,300	31.18	31,837
	–	36.91		40.66	
	↑	31.06		28.16	
CSI-2016	↓	38.34	143,262	25.99	30,000
	–	25.21		48.99	
	↑	36.45		25.02	

- 实验结果

Model	ACC %	F1 %
SVM [Tsantekidis *et al.*, 2017b]	-	49.42
MLP [Tsantekidis *et al.*, 2017b]	-	55.95
CNN-I [Tsantekidis *et al.*, 2017a]	-	59.44
LSTM [Tsantekidis *et al.*, 2017b]	-	61.43
CNN-II [Tsantekidis *et al.*, 2018]	-	47.00
B (TABL) [Tran *et al.*, 2018]	75.58	73.64
C (TABL) [Tran *et al.*, 2018]	79.87	78.44
DeepLOB [Zhang *et al.*, 2019]	80.51	80.35
BL-GAM-RHN-7 [Luo and Yu, 2019]	82.00	80.88
Downsampling RCNN	80.79	80.72
DWT XGBoost	80.81	80.74
MTDNN	81.12	81.05

Table 2: Results of predicting the mid-price movements in the next 50 events on FI-2010 dataset.

Model	ACC %	F1 %
SVM [Kim, 2003]	51.50	51.81
RF [Kara *et al.*, 2011]	52.30	51.96
TreNet [Lin *et al.*, 2017]	52.38	52.50
FDNN [Deng *et al.*, 2017]	52.32	52.45
CNNPred [Ehsan and Saman, 2019]	56.63	52.93
SFM [Hu and Qi, 2017]	52.96	52.97
DWT MLP [Lahmiri, 2014]	57.29	54.19
Downsampling RCNN	62.74	61.35
DWT XGBoost	62.19	60.74
MTDNN	63.07	61.65

Table 3: Results on CSI-2016.

- 消融实验

	Model	CSI-2016 ACC	CSI-2016 F1	FI-2010 ACC	FI-2010 F1
Single-scale	XGBoost	54.16	53.52	45.36	41.85
	CNN	56.63	52.93	56.73	58.22
	RNN	52.56	51.21	57.29	56.65
	RCNN	57.11	54.12	65.16	66.72
Multi-scale	DWT XGBoost	62.19	60.74	80.81	80.74
	DWT CNN	57.90	54.96	57.56	59.24
	DWT RNN	57.14	54.17	44.02	40.35
	DWT RCNN	57.58	50.21	44.58	37.71
	Downsampling XGBoost	54.18	53.51	45.60	41.82
	Downsampling RCNN*	57.21	51.95	59.45	58.03
	Downsampling RCNN	62.74	61.35	80.79	80.72
	Downsampling RCNN XGBoost	62.42	60.73	80.75	80.69
	Multi-kernel RCNN	58.14	54.40	67.16	66.85

Table 4: Ablation study